

TENDER SPECIFICATION

RECIRCULATING WATER CHILLER, 20 kW

1. Scope of Supply

Procurement of Re-circulating Chiller, 20 kW cooling capacity for laser cooling application at RRCAT, Indore.

Item No.	Category	Item Description	Key Specifications	Qty
1	Main Equipment	Re-circulating Chiller, 20 kW	Cooling = 20 kW @10°C	1 No
2	Accessories	Water hose pipe, insulated with nitrile rubber, with connectors to chiller	Pipe suitable to connect with chiller. 3 m length. Connector size =1"	2 Nos
3	Accessories	Quick connect coupling, 1" to 1"	1" BSP	2 Nos

2. Bidder Qualification Criteria (BQC)

2.1 Main Equipment Manufacturer: The OEM must have supplied at least 2 units of similar-capacity (20 kW or higher) re-circulating chillers to government / PSU / reputed organizations in the last 3 years. Supporting documents (Purchase Orders / Completion Certificates) must be provided as verifiable evidence. The OEM must hold valid ISO 9001:2015 and ISO 14001:2015 certifications. Copies of valid certificates must be enclosed with the bid.

2.2 Accessories / Subcomponents Sourcing: Only OEM or Authorized dealers are allowed to bid. The authorized dealers must submit OEM authorization certificate for the main chiller equipment on OEM letterhead.

3. Technical Requirements

3.1 Main Equipment

Parameter	Specification
Cooling Capacity	20 kW @ 10°C outlet temperature
Flow Rate	≥75 LPM (Stainless Steel 304 pump)
Pump Head Pressure	≥4 bar
Temperature Set Range	Ambient to 2°C
Temperature Control Accuracy	±1°C
Refrigerant	R404a / R-407C/ R-22 or equivalent
Compressor	Sealed type, energy efficient
Tank Capacity	≥60 L (SS 304)
Tank Insulation	25–50 mm Nitrile Rubber or PUF sheets
Material of Construction — Pump & Tank	SS 304 (corrosion resistant)
Material of Construction — Outer Frame	MS
Operating Environment	Forced air cooling, ambient ≤40°C
Electrical Supply	415V ±10%, three-phase, 50 Hz
Water Inlet / Outlet Coupling	1” BSP
Mains Cable	Heavy duty, 3-core, 5 m length
Portability	Heavy duty castor wheels (user removable) with locking arrangement
Controls	Digital temperature indicator cum controller
Filter	Dust filter mesh
Finish / Coating	Standard off-shelf finish

3.2 Accessories

Item	Specification
Water Hose Pipe (×2)	1” bore, 3 m length, nitrile rubber insulation on full length, with inlet/outlet connectors
Quick Connect Coupling (×2)	1” BSP to 1” BSP

3.3 Safety Features

Feature	Requirement
Water Low Level Trip	Automatic shutdown on low water level
Over Temperature Trip	Automatic shutdown on over-temperature
Overload Alarm	Alarm on compressor / pump overload
Trip Alarm	Visual / audible trip indication
Water Freezing Prevention	Protection against water freezing in cooling circuit

4. Site Acceptance Test

4.1 Testing: - The chiller working shall be demonstrated **by the supplier** at RRCAT, Indore upon delivery. All requirements mentioned in Section 3 shall be tested/demonstrated. Final acceptance and payment release shall be contingent on successful testing at RRCAT.

4.2 Documentation to be submitted with the bid:

- a) Duly filled and signed compliance sheet (Section 6 of this specification)
- b) Product catalogue with make & model specified
- c) OEM authorization certificate
- d) Past installations evidence.

Note: Make and model number of items shall be verifiable from the manufacturer's official website.

5. Delivery Terms

Term	Detail
Delivery Timeline	12 weeks from PO date
Packaging	Standard commercial packaging suitable for road transit
Warranty	12 months from date of final acceptance at RRCAT facility

6. Vendor Mandatory Compliance Sheet

Instructions: Bidders must indicate compliance clearly for every parameter. 'Yes/No/Complied' NOT ALLOWED. Attach supporting documents for each claim. Bids with incomplete or unsigned sheets may be summarily rejected.

No.	Parameter	Requirement	Supporting Document	Vendor Compliance
6.1	Make & Model Verification	Make & model number verifiable from manufacturer's website		
6.2	Quality Certifications	Valid ISO 9001:2015 + ISO 14001:2015		
6.3	OEM Authorization	OEM authorization certificate on OEM letterhead		
6.4	Cooling Capacity	20 kW @10°C		
6.5	Flow Rate	≥75 LPM (SS 304 pump)		
6.6	Minimum Temperature	2°C		
6.6	Pump Head Pressure	≥4 bar		
6.7	Tank Capacity	≥60 L, SS 304		
6.8	Refrigerant	R404a / R-407C / R-22 or equivalent		
6.9	Electrical Supply	415V ±10%, 3-phase, 50 Hz		
6.10	Pipe	Water hose pipes		
6.11	Coupling	Quick connect coupling		

Signature of Bidder: _____

Company Seal: _____

Date: _____